IMPORT THE REQUIRED PYTHON LIBRARIES

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

IMPORT THE FLIGHTS DATASET

```
data_path = '/kaggle/input/flight-delay/'
flights = pd.read_csv(data_path + 'flights.csv')
```

```
flights.shape
```

```
(5819079, 31)
```

```
flights.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5819079 entries, 0 to 5819078
Data columns (total 31):
 #   Column              Dtype  
---  ------              -----  
 0   YEAR                int64  
 1   MONTH               int64  
 2   DAY                 int64  
 3   DAY_OF_WEEK         int64  
 4   AIRLINE             object 
 5   FLIGHT_NUMBER       int64  
 6   TAIL_NUMBER         object 
 7   ORIGIN_AIRPORT      object 
 8   DESTINATION_AIRPORT object 
 9   SCHEDULED_DEPARTURE int64  
 10  DEPARTURE_TIME      float64
 11  DEPARTURE_DELAY     float64
 12  TAXI_OUT            float64
 13  WHEELS_OFF          float64
 14  SCHEDULED_TIME      float64
 15  ELAPSED_TIME        float64
 16  AIR_TIME            float64
 17  DISTANCE            int64  
 18  WHEELS_ON           float64
 19  TAXI_IN             float64
 20  SCHEDULED_ARRIVAL   int64  
 21  ARRIVAL_TIME        float64
 22  ARRIVAL_DELAY       float64
 23  DIVERTED            int64  
 24  CANCELLED           int64  
 25  CANCELLATION_REASON object 
 26  AIR_SYSTEM_DELAY    float64
 27  SECURITY_DELAY      float64
 28  AIRLINE_DELAY       float64
 29  LATE_AIRCRAFT_DELAY float64
 30  WEATHER_DELAY       float64
dtypes: float64(16), int64(11), object(4)
memory usage: 1.3 GB
```

```
data.head()
```

	YEAR	MONTH	DAY	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	DEPARTURE_TIME	DEPARTURE_DELAY	TAXI_OUT	WHEELS_OFF	SCHEDULED_TIME	ELAPSED_TIME	AIR_TIME	DISTANCE	WHEELS_ON	TAXI_IN	SCHEDULED_ARRIVAL	ARRIVAL_TIME	ARRIVAL_DELAY	DIVERTED	CANCELLED	CANCELLATION_REASON	AIR_SYSTEM_DELAY	SECURITY_DELAY	AIRLINE_DELAY	LATE_AIRCRAFT_DELAY	WEATHER_DELAY
0	2015	1	1	4	AS	98	N407AS	ANC	SEA	0005	NaN	NaN	NaN	NaN	205	NaN	NaN	1448	NaN	NaN	0820	NaN	NaN	0	1	B	NaN	NaN	NaN	NaN	NaN
1	2015	1	1	4	AA	2336	N3KUAA	LAX	PBI	0010	0448	-30.0	12.0	0500	337	318.0	293.0	2330	0518.0	13.0	0547	0531.0	-16.0	0	0		NaN	NaN	NaN	NaN	NaN
2	2015	1	1	4	US	840	N171US	SFO	CLT	0020	0133	-47.0	18.0	0151	277	256.0	228.0	1935	0547.0	10.0	0557	0557.0	0.0	0	0		NaN	NaN	NaN	NaN	NaN
3	2015	1	1	4	AA	258	N3HYAA	LAX	MIA	0020	0324	64.0	7.0	0331	351	326.0	299.0	2342	0630.0	20.0	0651	0650.0	-1.0	0	0		0.0	0.0	0.0	64.0	0.0
4	2015	1	1	4	AS	135	N527AS	SEA	ANC	0025	0455	30.0	22.0	0517	157	142.0	114.0	1448	0649.0	6.0	0702	0655.0	-7.0	0	0		NaN	NaN	NaN	NaN	NaN

```
data.tail()
```

	YEAR	MONTH	DAY	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	DEPARTURE_TIME	DEPARTURE_DELAY	TAXI_OUT	WHEELS_OFF	SCHEDULED_TIME	ELAPSED_TIME	AIR_TIME	DISTANCE	WHEELS_ON	TAXI_IN	SCHEDULED_ARRIVAL	ARRIVAL_TIME	ARRIVAL_DELAY	DIVERTED	CANCELLED	CANCELLATION_REASON	AIR_SYSTEM_DELAY	SECURITY_DELAY	AIRLINE_DELAY	LATE_AIRCRAFT_DELAY	WEATHER_DELAY
5819074	2015	12	31	4	B6	1537	N516JB	BOS	FLL	2350	2358.0	8.0	17.0	0015	189	161.0	137.0	1121	0232.0	7.0	0259	0239.0	-20.0	0	0		NaN	NaN	NaN	NaN	NaN
5819075	2015	12	31	4	B6	1538	N804JB	BQN	BOS	2359	0000.0	1.0	14.0	0014	224	201.0	187.0	1536	0221.0	6.0	0243	0227.0	-16.0	0	0		NaN	NaN	NaN	NaN	NaN
5819076	2015	12	31	4	B6	1539	N623JB	MCO	SJU	2359	0000.0	1.0	15.0	0015	193	171.0	151.0	1074	0246.0	5.0	0252	0251.0	-1.0	0	0		NaN	NaN	NaN	NaN	NaN
5819077	2015	12	31	4	B6	1540	N635JB	FLL	BQN	2359	0000.0	1.0	16.0	0016	198	174.0	154.0	1176	0250.0	4.0	0257	0254.0	-3.0	0	0		NaN	NaN	NaN	NaN	NaN
5819078	2015	12	31	4	B6	1541	N794JB	LAS	BOS	2359	0000.0	1.0	13.0	0013	291	263.0	242.0	2172	0515.0	8.0	0510	0523.0	13.0	0	0		NaN	NaN	NaN	NaN	NaN

```
data.columns
```

```
Index(['YEAR', 'MONTH', 'DAY', 'DAY_OF_WEEK', 'AIRLINE', 'FLIGHT_NUMBER',
       'TAIL_NUMBER', 'ORIGIN_AIRPORT', 'DESTINATION_AIRPORT',
       'SCHEDULED_DEPARTURE', 'DEPARTURE_TIME', 'DEPARTURE_DELAY', 'TAXI_OUT',
       'WHEELS_OFF', 'SCHEDULED_TIME', 'ELAPSED_TIME', 'AIR_TIME',
       'DISTANCE', 'WHEELS_ON', 'TAXI_IN', 'SCHEDULED_ARRIVAL',
       'ARRIVAL_TIME', 'ARRIVAL_DELAY', 'DIVERTED', 'CANCELLED',
       'CANCELLATION_REASON', 'AIR_SYSTEM_DELAY', 'SECURITY_DELAY',
       'AIRLINE_DELAY', 'LATE_AIRCRAFT_DELAY', 'WEATHER_DELAY'],
      dtype='object')
```

CLEANING THE DATA

```
data.drop(columns = ['YEAR', 'CANCELLED'], inplace = True)
```

```
data.head()
```

	MONTH	DAY	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	DEPARTURE_TIME	DEPARTURE_DELAY	TAXI_OUT	WHEELS_OFF	SCHEDULED_TIME	ELAPSED_TIME	AIR_TIME	DISTANCE	WHEELS_ON	TAXI_IN	SCHEDULED_ARRIVAL	ARRIVAL_TIME	ARRIVAL_DELAY	DIVERTED	CANCELLATION_REASON	AIR_SYSTEM_DELAY	SECURITY_DELAY	AIRLINE_DELAY	LATE_AIRCRAFT_DELAY	WEATHER_DELAY
0	1	1	4	AS	98	N407AS	ANC	SEA	0005	NaN	NaN	NaN	NaN	205	NaN	NaN	1448	NaN	NaN	0820	NaN	NaN	0	B	NaN	NaN	NaN	NaN	NaN	NaN
1	1	1	4	AA	2336	N3KUAA	LAX	PBI	0010	0448	-30.0	12.0	0500	337	318.0	293.0	2330	0518.0	13.0	0547	0531.0	-16.0	0		NaN	NaN	NaN	NaN	NaN	NaN
2	1	1	4	US	840	N171US	SFO	CLT	0020	0133	-47.0	18.0	0151	277	256.0	228.0	1935	0547.0	10.0	0557	0557.0	0.0	0		NaN	NaN	NaN	NaN	NaN	NaN
3	1	1	4	AA	258	N3HYAA	LAX	MIA	0020	0324	64.0	7.0	0331	351	326.0	299.0	2342	0630.0	20.0	0651	0650.0	-1.0	0		0.0	0.0	0.0	64.0	0.0	NaN
4	1	1	4	AS	135	N527AS	SEA	ANC	0025	0455	30.0	22.0	0517	157	142.0	114.0	1448	0649.0	6.0	0702	0655.0	-7.0	0		NaN	NaN	NaN	NaN	NaN	NaN

```
data.tail()
```

	MONTH	DAY	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	DEPARTURE_TIME	DEPARTURE_DELAY	TAXI_OUT	WHEELS_OFF	SCHEDULED_TIME	ELAPSED_TIME	AIR_TIME	DISTANCE	WHEELS_ON	TAXI_IN	SCHEDULED_ARRIVAL	ARRIVAL_TIME	ARRIVAL_DELAY	DIVERTED	CANCELLATION_REASON	AIR_SYSTEM_DELAY	SECURITY_DELAY	AIRLINE_DELAY	LATE_AIRCRAFT_DELAY	WEATHER_DELAY
5819074	12	31	4	B6	1537	N516JB	BOS	FLL	2350	2358.0	8.0	17.0	0015	189	161.0	137.0	1121	0232.0	7.0	0259	0239.0	-20.0	0		NaN	NaN	NaN	NaN	NaN
5819075	12	31	4	B6	1538	N804JB	BQN	BOS	2359	0000.0	1.0	14.0	0014	224	201.0	187.0	1536	0221.0	6.0	0243	0227.0	-16.0	0		NaN	NaN	NaN	NaN	NaN
5819076	12	31	4	B6	1539	N623JB	MCO	SJU	2359	0000.0	1.0	15.0	0015	193	171.0	151.0	1074	0246.0	5.0	0252	0251.0	-1.0	0		NaN	NaN	NaN	NaN	NaN
5819077	12	31	4	B6	1540	N635JB	FLL	BQN	2359	0000.0	1.0	16.0	0016	198	174.0	154.0	1176	0250.0	4.0	0257	0254.0	-3.0	0		NaN	NaN	NaN	NaN	NaN
5819078	12	31	4	B6	1541	N794JB	LAS	BOS	2359	0000.0	1.0	13.0	0013	291	263.0	242.0	2172	0515.0	8.0	0510	0523.0	13.0	0		NaN	NaN	NaN	NaN	NaN

```
data.shape
```

```
(5819079, 29)
```

```
data.isnull().sum()
```

```
MONTH                       0
DAY                         0
DAY_OF_WEEK                 0
AIRLINE                     0
FLIGHT_NUMBER               0
TAIL_NUMBER                 0
ORIGIN_AIRPORT              0
DESTINATION_AIRPORT         0
SCHEDULED_DEPARTURE         0
DEPARTURE_TIME         86153
DEPARTURE_DELAY        86153
TAXI_OUT               89047
WHEELS_OFF             86153
SCHEDULED_TIME             6
ELAPSED_TIME           105071
AIR_TIME               105071
DISTANCE                    0
WHEELS_ON              86153
TAXI_IN                89047
SCHEDULED_ARRIVAL           0
ARRIVAL_TIME           86153
ARRIVAL_DELAY          105071
DIVERTED                    0
CANCELLATION_REASON    5729167
AIR_SYSTEM_DELAY       4755640
SECURITY_DELAY         4755640
AIRLINE_DELAY          4755640
LATE_AIRCRAFT_DELAY    4755640
WEATHER_DELAY          4755640
dtype: int64
```

```
data.dtypes
```

```
MONTH                  int64
DAY                    int64
DAY_OF_WEEK            int64
AIRLINE               object
FLIGHT_NUMBER          int64
TAIL_NUMBER           object
ORIGIN_AIRPORT        object
DESTINATION_AIRPORT   object
SCHEDULED_DEPARTURE    int64
DEPARTURE_TIME       float64
DEPARTURE_DELAY      float64
TAXI_OUT             float64
WHEELS_OFF           float64
SCHEDULED_TIME       float64
ELAPSED_TIME         float64
AIR_TIME             float64
DISTANCE               int64
WHEELS_ON            float64
TAXI_IN              float64
SCHEDULED_ARRIVAL      int64
ARRIVAL_TIME         float64
ARRIVAL_DELAY        float64
DIVERTED               int64
CANCELLATION_REASON   object
AIR_SYSTEM_DELAY     float64
SECURITY_DELAY       float64
AIRLINE_DELAY        float64
LATE_AIRCRAFT_DELAY  float64
WEATHER_DELAY        float64
dtype: int64
```

```
data.groupby('CANCELLATION_REASON').count()
```

CANCELLATION_REASON	MONTH	DAY	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	DEPARTURE_TIME	DEPARTURE_DELAY	TAXI_OUT	WHEELS_OFF	SCHEDULED_TIME	ELAPSED_TIME	AIR_TIME	DISTANCE	WHEELS_ON	TAXI_IN	SCHEDULED_ARRIVAL	ARRIVAL_TIME	ARRIVAL_DELAY	DIVERTED	AIR_SYSTEM_DELAY	SECURITY_DELAY	AIRLINE_DELAY	LATE_AIRCRAFT_DELAY	WEATHER_DELAY
A	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851	48851
B	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522	62522
C	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790	19790
D	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548	5548

```
data.groupby('DIVERTED').count()
```

DIVERTED	MONTH	DAY	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	DEPARTURE_TIME	DEPARTURE_DELAY	TAXI_OUT	WHEELS_OFF	SCHEDULED_TIME	ELAPSED_TIME	AIR_TIME	DISTANCE	WHEELS_ON	TAXI_IN	SCHEDULED_ARRIVAL	ARRIVAL_TIME	ARRIVAL_DELAY	CANCELLATION_REASON	AIR_SYSTEM_DELAY	SECURITY_DELAY	AIRLINE_DELAY	LATE_AIRCRAFT_DELAY	WEATHER_DELAY
0	5732926	5732926	5732926	5732926	5732926	5732926	5732926	5732926	5732926	5646773	5646773	5643879	5646773	5732920	5627855	5627855	5732926	5646773	5643879	5732926	5646773	5627855	5729167	1064286	1064286	1064286	1064286	1064286
1	86153	86153	86153	86153	86153	86153	86153	86153	86153	0	0	0	0	86147	0	0	86153	0	0	86153	0	0	86153	3691354	3691354	3691354	3691354	3691354

```
data.groupby('DIVERTED').count.shape
```

```
(2, 29)
```

```
data.groupby('DIVERTED').count.T
```

DIVERTED	0	1
MONTH	5732926	86153
DAY	5732926	86153
DAY_OF_WEEK	5732926	86153
AIRLINE	5732926	86153
FLIGHT_NUMBER	5732926	86153
TAIL_NUMBER	5732926	86153
ORIGIN_AIRPORT	5732926	86153
DESTINATION_AIRPORT	5732926	86153
SCHEDULED_DEPARTURE	5732926	86153
DEPARTURE_TIME	5646773	0
DEPARTURE_DELAY	5646773	0
TAXI_OUT	5643879	0
WHEELS_OFF	5646773	0
SCHEDULED_TIME	5732920	86147
ELAPSED_TIME	5627855	0
AIR_TIME	5627855	0
DISTANCE	5732926	86153
WHEELS_ON	5646773	0
TAXI_IN	5643879	0
SCHEDULED_ARRIVAL	5732926	86153
ARRIVAL_TIME	5646773	0
ARRIVAL_DELAY	5627855	0
CANCELLATION_REASON	5729167	3691354
AIR_SYSTEM_DELAY	1064286	3691354
SECURITY_DELAY	1064286	3691354
AIRLINE_DELAY	1064286	3691354
LATE_AIRCRAFT_DELAY	1064286	3691354
WEATHER_DELAY	1064286	3691354

```
data.groupby('CANCELLATION_REASON').count.T
```

CANCELLATION_REASON	A	B	C	D
MONTH	48851	62522	19790	5548
DAY	48851	62522	19790	5548
DAY_OF_WEEK	48851	62522	19790	5548
AIRLINE	48851	62522	19790	5548
FLIGHT_NUMBER	48851	62522	19790	5548
TAIL_NUMBER	48851	62522	19790	5548
ORIGIN_AIRPORT	48851	62522	19790	5548
DESTINATION_AIRPORT	48851	62522	19790	5548
SCHEDULED_DEPARTURE	48851	62522	19790	5548
DEPARTURE_TIME	48851	62522	19790	5548
DEPARTURE_DELAY	48851	62522	19790	5548
TAXI_OUT	48851	62522	19790	5548
WHEELS_OFF	48851	62522	19790	5548
SCHEDULED_TIME	48851	62522	19790	5548
ELAPSED_TIME	48851	62522	19790	5548
AIR_TIME	48851	62522	19790	5548
DISTANCE	48851	62522	19790	5548
WHEELS_ON	48851	62522	19790	5548
TAXI_IN	48851	62522	19790	5548
SCHEDULED_ARRIVAL	48851	62522	19790	5548
ARRIVAL_TIME	48851	62522	19790	5548
ARRIVAL_DELAY	48851	62522	19790	5548
DIVERTED	48851	62522	19790	5548
AIR_SYSTEM_DELAY	48851	62522	19790	5548
SECURITY_DELAY	48851	62522	19790	5548
AIRLINE_DELAY	48851	62522	19790	5548
LATE_AIRCRAFT_DELAY	48851	62522	19790	5548
WEATHER_DELAY	48851	62522	19790	5548

```
data.groupby('CANCELLATION_REASON').count.shape
```

```
(4, 29)
```

```
data.groupby('CANCELLATION_REASON').count.T.shape
```

```
(29, 4)
```

```
data.groupby('CANCELLATION_REASON').count.T.sum()
```

```
MONTH                  5729167
DAY                    5729167
DAY_OF_WEEK            5729167
AIRLINE                5729167
FLIGHT_NUMBER          5729167
TAIL_NUMBER            5729167
ORIGIN_AIRPORT         5729167
DESTINATION_AIRPORT    5729167
SCHEDULED_DEPARTURE    5729167
DEPARTURE_TIME         5729167
DEPARTURE_DELAY        5729167
TAXI_OUT               5729167
WHEELS_OFF             5729167
SCHEDULED_TIME         5729167
ELAPSED_TIME           5729167
AIR_TIME               5729167
DISTANCE               5729167
WHEELS_ON              5729167
TAXI_IN                5729167
SCHEDULED_ARRIVAL      5729167
ARRIVAL_TIME           5729167
ARRIVAL_DELAY          5729167
DIVERTED               5729167
AIR_SYSTEM_DELAY       1064286
SECURITY_DELAY         1064286
AIRLINE_DELAY          1064286
LATE_AIRCRAFT_DELAY    1064286
WEATHER_DELAY          1064286
dtype: int64
```

WHAT ARE THE AIRLINES IN THE DATASET ACCOUNTAMED BY THEIR FREQUENCIES?

```
data.groupby('AIRLINE').count.T
```

```
data.groupby('AIRLINE').count.T.shape
```

```
(29, 14)
```

WHICH ORIGIN AIRPORT HAS THE HIGHEST DEPARTURE DELAYS?

```
data.groupby('ORIGIN_AIRPORT').count.T
```

```
data.groupby('ORIGIN_AIRPORT').count.T.shape
```

```
(29, 322)
```

```
data.groupby('ORIGIN_AIRPORT').count.T.sum()
```

```
MONTH                  5732926
DAY                    5732926
DAY_OF_WEEK            5732926
AIRLINE                5732926
FLIGHT_NUMBER          5732926
TAIL_NUMBER            5732926
ORIGIN_AIRPORT         5732926
DESTINATION_AIRPORT    5732926
SCHEDULED_DEPARTURE    5732926
DEPARTURE_TIME         5646773
DEPARTURE_DELAY        5646773
TAXI_OUT               5643879
WHEELS_OFF             5646773
SCHEDULED_TIME         5732920
ELAPSED_TIME           5627855
AIR_TIME               5627855
DISTANCE               5732926
WHEELS_ON              5646773
TAXI_IN                5643879
SCHEDULED_ARRIVAL      5732926
ARRIVAL_TIME           5646773
ARRIVAL_DELAY          5627855
DIVERTED               5732926
CANCELLATION_REASON    5729167
AIR_SYSTEM_DELAY       1064286
SECURITY_DELAY         1064286
AIRLINE_DELAY          1064286
LATE_AIRCRAFT_DELAY    1064286
WEATHER_DELAY          1064286
dtype: int64
```

```
data.groupby('ORIGIN_AIRPORT').count.T.sum()
```

```
MONTH                  5732926
DAY                    5732926
DAY_OF_WEEK            5732926
AIRLINE                5732926
FLIGHT_NUMBER          5732926
TAIL_NUMBER            5732926
ORIGIN_AIRPORT         5732926
DESTINATION_AIRPORT    5732926
SCHEDULED_DEPARTURE    5732926
DEPARTURE_TIME         5646773
DEPARTURE_DELAY        5646773
TAXI_OUT               5643879
WHEELS_OFF             5646773
SCHEDULED_TIME         5732920
ELAPSED_TIME           5627855
AIR_TIME               5627855
DISTANCE               5732926
WHEELS_ON              5646773
TAXI_IN                5643879
SCHEDULED_ARRIVAL      5732926
ARRIVAL_TIME           5646773
ARRIVAL_DELAY          5627855
DIVERTED               5732926
CANCELLATION_REASON    5729167
AIR_SYSTEM_DELAY       1064286
SECURITY_DELAY         1064286
AIRLINE_DELAY          1064286
LATE_AIRCRAFT_DELAY    1064286
WEATHER_DELAY          1064286
dtype: int64
```

```
data.groupby('ORIGIN_AIRPORT').count.T.shape
```

```
(29, 322)
```

```
data.groupby('ORIGIN_AIRPORT').count.T.sum.shape
```

```
(29,)
```

```
data.groupby('ORIGIN_AIRPORT').count.T.sum.T
```

```
MONTH                  5732926
DAY                    5732926
DAY_OF_WEEK            5732926
AIRLINE                5732926
FLIGHT_NUMBER          5732926
TAIL_NUMBER            5732926
ORIGIN_AIRPORT         5732926
DESTINATION_AIRPORT    5732926
SCHEDULED_DEPARTURE    5732926
DEPARTURE_TIME         5646773
DEPARTURE_DELAY        5646773
TAXI_OUT               5643879
WHEELS_OFF             5646773
SCHEDULED_TIME         5732920
ELAPSED_TIME           5627855
AIR_TIME               5627855
DISTANCE               5732926
WHEELS_ON              5646773
TAXI_IN                5643879
SCHEDULED_ARRIVAL      5732926
ARRIVAL_TIME           5646773
ARRIVAL_DELAY          5627855
DIVERTED               5732926
CANCELLATION_REASON    5729167
AIR_SYSTEM_DELAY       1064286
SECURITY_DELAY         1064286
AIRLINE_DELAY          1064286
LATE_AIRCRAFT_DELAY    1064286
WEATHER_DELAY          1064286
dtype: int64
```

```
data.groupby('ORIGIN_AIRPORT').count.T.sum.T.shape
```

```
(29,)
```

```
data.groupby('ORIGIN_AIRPORT').count.T.sum.T.sum()
```

```
166143160
```

```
data.groupby('ORIGIN_AIRPORT').count.T.sum.T.sum().shape
```

```
()
```

WHICH DESTINATION AIRPORT HAS THE HIGHEST ARRIVAL DELAYS?

```
data.groupby('DESTINATION_AIRPORT').count.T
```

```
data.groupby('DESTINATION_AIRPORT').count.T.shape
```

```
(29, 322)
```

```
data.groupby('DESTINATION_AIRPORT').count.T.sum()
```

```
MONTH                  5732926
DAY                    5732926
DAY_OF_WEEK            5732926
AIRLINE                5732926
FLIGHT_NUMBER          5732926
TAIL_NUMBER            5732926
ORIGIN_AIRPORT         5732926
DESTINATION_AIRPORT    5732926
SCHEDULED_DEPARTURE    5732926
DEPARTURE_TIME         5646773
DEPARTURE_DELAY        5646773
TAXI_OUT               5643879
WHEELS_OFF             5646773
SCHEDULED_TIME         5732920
ELAPSED_TIME           5627855
AIR_TIME               5627855
DISTANCE               5732926
WHEELS_ON              5646773
TAXI_IN                5643879
SCHEDULED_ARRIVAL      5732926
ARRIVAL_TIME           5646773
ARRIVAL_DELAY          5627855
DIVERTED               5732926
CANCELLATION_REASON    5729167
AIR_SYSTEM_DELAY       1064286
SECURITY_DELAY         1064286
AIRLINE_DELAY          1064286
LATE_AIRCRAFT_DELAY    1064286
WEATHER_DELAY          1064286
dtype: int64
```

```
data.groupby('DESTINATION_AIRPORT').count.T.sum
```

```
MONTH                  5732926
DAY                    5732926
DAY_OF_WEEK            5732926
AIRLINE                5732926
FLIGHT_NUMBER          5732926
TAIL_NUMBER            5732926
ORIGIN_AIRPORT         5732926
DESTINATION_AIRPORT    5732926
SCHEDULED_DEPARTURE    5732926
DEPARTURE_TIME         5646773
DEPARTURE_DELAY        5646773
TAXI_OUT               5643879
WHEELS_OFF             5646773
SCHEDULED_TIME         5732920
ELAPSED_TIME           5627855
AIR_TIME               5627855
DISTANCE               5732926
WHEELS_ON              5646773
TAXI_IN                5643879
SCHEDULED_ARRIVAL      5732926
ARRIVAL_TIME           5646773
ARRIVAL_DELAY          5627855
DIVERTED               5732926
CANCELLATION_REASON    5729167
AIR_SYSTEM_DELAY       1064286
SECURITY_DELAY         1064286
AIRLINE_DELAY          1064286
LATE_AIRCRAFT_DELAY    1064286
WEATHER_DELAY          1064286
dtype: int64
```

```
data.groupby('DESTINATION_AIRPORT').count.T.sum.shape
```

```
(29,)
```

```
data.groupby('DESTINATION_AIRPORT').count.T.sum.T
```

```
MONTH                  5732926
DAY                    5732926
DAY_OF_WEEK            5732926
AIRLINE                5732926
FLIGHT_NUMBER          5732926
TAIL_NUMBER            5732926
ORIGIN_AIRPORT         5732926
DESTINATION_AIRPORT    5732926
SCHEDULED_DEPARTURE    5732926
DEPARTURE_TIME         5646773
DEPARTURE_DELAY        5646773
TAXI_OUT               5643879
WHEELS_OFF             5646773
SCHEDULED_TIME         5732920
ELAPSED_TIME           5627855
AIR_TIME               5627855
DISTANCE               5732926
WHEELS_ON              5646773
TAXI_IN                5643879
SCHEDULED_ARRIVAL      5732926
ARRIVAL_TIME           5646773
ARRIVAL_DELAY          5627855
DIVERTED               5732926
CANCELLATION_REASON    5729167
AIR_SYSTEM_DELAY       1064286
SECURITY_DELAY         1064286
AIRLINE_DELAY          1064286
LATE_AIRCRAFT_DELAY    1064286
WEATHER_DELAY          1064286
dtype: int64
```

```
data.groupby('DESTINATION_AIRPORT').count.T.sum.T.shape
```

```
(29,)
```

```
data.groupby('DESTINATION_AIRPORT').count.T.sum.T.sum()
```

```
166143160
```

```
data.groupby('DESTINATION_AIRPORT').count.T.sum.T.sum().shape
```

```
()
```

WHICH DAY OF THE WEEK IS THE MOST DELAYED DAY?

```
data.groupby('DAY_OF_WEEK').count.T
```

```
data.groupby('DAY_OF_WEEK').count.T.shape
```

```
(29, 7)
```

```
data.groupby('DAY_OF_WEEK').count.T.sum
```

```
MONTH                  5732926
DAY                    5732926
DAY_OF_WEEK            5732926
AIRLINE                5732926
FLIGHT_NUMBER          5732926
TAIL_NUMBER            5732926
ORIGIN_AIRPORT         5732926
DESTINATION_AIRPORT    5732926
SCHEDULED_DEPARTURE    5732926
DEPARTURE_TIME         5646773
DEPARTURE_DELAY        5646773
TAXI_OUT               5643879
WHEELS_OFF             5646773
SCHEDULED_TIME         5732920
ELAPSED_TIME           5627855
AIR_TIME               5627855
DISTANCE               5732926
WHEELS_ON              5646773
TAXI_IN                5643879
SCHEDULED_ARRIVAL      5732926
ARRIVAL_TIME           5646773
ARRIVAL_DELAY          5627855
DIVERTED               5732926
CANCELLATION_REASON    5729167
AIR_SYSTEM_DELAY       1064286
SECURITY_DELAY         1064286
AIRLINE_DELAY          1064286
LATE_AIRCRAFT_DELAY    1064286
WEATHER_DELAY          1064286
dtype: int64
```

```
data.groupby('DAY_OF_WEEK').count.T.sum.shape
```

```
(29,)
```

```
data.groupby('DAY_OF_WEEK').count.T.sum.T
```

```
MONTH                  5732926
DAY                    5732926
DAY_OF_WEEK            5732926
AIRLINE                5732926
FLIGHT_NUMBER          5732926
TAIL_NUMBER            5732926
ORIGIN_AIRPORT         5732926
DESTINATION_AIRPORT    5732926
SCHEDULED_DEPARTURE    5732926
DEPARTURE_TIME         5646773
DEPARTURE_DELAY        5646773
TAXI_OUT               5643879
WHEELS_OFF             5646773
SCHEDULED_TIME         5732920
ELAPSED_TIME           5627855
AIR_TIME               5627855
DISTANCE               5732926
WHEELS_ON              5646773
TAXI_IN                5643879
SCHEDULED_ARRIVAL      5732926
ARRIVAL_TIME           5646773
ARRIVAL_DELAY          5627855
DIVERTED               5732926
CANCELLATION_REASON    5729167
AIR_SYSTEM_DELAY       1064286
SECURITY_DELAY         1064286
AIRLINE_DELAY          1064286
LATE_AIRCRAFT_DELAY    1064286
WEATHER_DELAY          1064286
dtype: int64
```

```
data.groupby('DAY_OF_WEEK').count.T.sum.T.shape
```

```
(29,)
```

```
data.groupby('DAY_OF_WEEK').count.T.sum.T.sum()
```

```
166143160
```

```
data.groupby('DAY_OF_WEEK').count.T.sum.T.sum().shape
```

```
()
```

WHICH AIRLINE HAS THE HIGHEST DEPARTURE DELAYS?

```
data.groupby('AIRLINE').count.T
```

```
data.groupby('AIRLINE').count.T.shape
```

```
(29, 14)
```

```
data.groupby('AIRLINE').count.T.sum
```

```
MONTH                  5732926
DAY                    5732926
DAY_OF_WEEK            5732926
AIRLINE                5732926
FLIGHT_NUMBER          5732926
TAIL_NUMBER            5732926
ORIGIN_AIRPORT         5732926
DESTINATION_AIRPORT    5732926
SCHEDULED_DEPARTURE    5732926
DEPARTURE_TIME         5646773
DEPARTURE_DELAY        5646773
TAXI_OUT               5643879
WHEELS_OFF             5646773
SCHEDULED_TIME
```

```
Array("Mercedes", "Porsche", "Ferrari", "BLAUPUNKT", "Tomtom",
      "Bosch", "Brembo", "Pirelli")
```

```
Element.$.map(Array(Math.floor(Math.random() * (Array.length-1)), Math.floor(Math.random() *
(Array.length-1))), function(i, j) {
  return Array[i] + " " + Array[j];
});
```

```
Element.$.text("Hello");
```

```
$('.slide-0').fadeIn.delay(400).fadeOut(500);
```